JUAN LOPES

SOFTWARE ENGINEER

Rio de Janeiro, Brazil me@juanlopes.net juanlopes.net github.com/juanplopes November 21, 2025

EXPERIENCE

Sr Staff Software Engineer, Game Server

Zwift

Jun 2020 – Ongoing (5 yr 5 mos) Rio de Janeiro / Long Beach

- Rewrote the spatial index data structure, increasing the throughput of k-nearest neighbors queries in more than 800%.
- Implemented locality-based UDP partitioning strategy, allowing them to handle more than 300,000 concurrent users.
- Optimized low-latency server algorithms, increasing the per-server capacity in more than 500%.
- Created the main Game automation tool used by more than 100 engineers in the company.

Lead Software Engineer, R&D

Intelie

Aug 2011 – May 2020 (8 yrs 11 mos) Rio de Janeiro / Houston

- Responsible for the Core Platform roadmap and implementation.
- Created Intelie Pipes, a taylor-built distributed stream processing engine (**pipes.intelie.com**).

Software Engineer

Living Consultoria

Dec 2007 – Aug 2011 (3 yrs 8 mos) Rio de Janeiro

- Delivered projects for several customers around the globe, with great focus on billing solutions;

EDUCATION

Ph.D. in Systems Engineering and Computer Science

Federal University of Rio de Janeiro

Jun 2017 – Jan 2020, GPA: 4.0 – **Incomplete, ABD**

Completed regular course, but did not finish the dissertation. Researched probabilistic streaming graph representations.

M.Sc. in Computational Sciences

State University of Rio de Janeiro

Mar 2014 – Mar 2017, GPA: 3.8

Researched probabilistic data structures applied to implicit graph representation

B.Sc. in Informatics and Information Technology

State University of Rio de Janeiro

Jan 2006 – Aug 2013, GPA: 3.0

MOST INTERESTED IN

I love algorithms & data structures, software optimization, and compilers. I enjoy bridging the gap between theoretical and applied computer science through code.

MOST PROUD OF

Master's Thesis Award
The master's thesis "Probabilistic data structures applied to implicit graph representation" was chosen as one of the top 9 theses of 2017 by the Brazilian Computer Society.
sol.sbc.org.br/index.php/ctd/issue/view/233

StackOverflow Rep. in [algorithm]
I have a 10k+ reputation on StackOverflow, which is not a huge number, but this still makes me proud, given I mostly only answer questions in the not-so-popular [algorithm] tag.
stackoverflow.com/users/1327235

Competitive Programming
I really enjoy programming contests. In the IEEEXtreme, I have once won a 1st place in Brazil (2013) and 34th place worldwide (2014). In the ICPC, I was five times South-American finalist. My most popular GitHub repository is a collection of problems solved in my early training years.
github.com/juanplopes/icpc

QCon SP Program Committee
QCon is a prestigious international software conference. After many years as a regular speaker, in 2018, I was invited to be part of the Program Committee, hosting the Computer Science track in São Paulo edition.
qconsp.com/sp2020/track/ciencia-da-computacao-no-mundo-real

Other work
I have recently been fiddling with JVM introspection and heap overhead.
github.com/intelie/introspective
Using that library, I have managed to create an aggressively space optimized implementation of java.util.Map:
github.com/intelie/tinymap